DO NOW · AO1 REVISION

Distance, displacement, speed and velocity

Forces
Paper 2 · Combined and Separate
15 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 6 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**Define distance.**	How far an object moves.	☐
2	**Define displacement.**	The straight-line distance from start to finish, with its direction.	☐
3 ★	**Explain the difference between distance and displacement.**	Displacement has a direction (vector); distance does not (scalar).	☐
4 ★	**Explain the difference between speed and velocity.**	Velocity has a direction (vector); speed does not (scalar).	☐
5 ★	**State typical values for walking speed, running speed and cycling speed.**	Walking: 1.5 m/s Running: 3 m/s Cycling: 6 m/s	☐
6	**State a typical value for the speed of sound in air.**	330 m/s	☐
7 ★	**Define velocity.**	The speed of an object in a given direction.	☐
8	**An object moves along a curved path. Describe its speed and its velocity.**	Speed: can be constant. Velocity: changing, because the direction changes.	☐
9 ★	**State what the gradient of a distance–time graph represents.**	Speed	☐
10	**Describe how acceleration appears on a distance–time graph.**	As a curve (the gradient changes).	☐
11	**State the equation linking distance travelled, speed and time.**	distance travelled = speed × time ($s = v\ t$)	☐
12	**Give three factors that affect how fast a person walks, runs or cycles.**	Any three from: age, fitness, terrain, distance travelled	☐

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	QUESTION	ANSWER	✓
13	**State typical speeds for a car on a main road, a train and an aeroplane.**	Car: about 25 m/s Train: about 50 m/s Aeroplane: about 250 m/s	☐
14	**State what a horizontal line on a distance–time graph shows.**	The object is stationary.	☐
15 ★	**Describe how to find the speed at one instant from a curved distance–time graph.**	Draw a tangent to the curve at that time and find its gradient.	☐

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